

Solution of Algebraic and Transcendental Equations

Lab Manual 01

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1 Bisection Method

1.1 Brief Description

The Bisection Method is a reliable numerical technique used to find a root of a continuous function $f(x)$ in a given interval $[a, b]$ where the function changes sign, i.e., $f(a) \cdot f(b) < 0$. The method works by repeatedly bisecting the interval and selecting the subinterval in which the sign change occurs. This process is repeated until the interval is sufficiently small or the function value at the midpoint is close enough to zero.

1.2 Procedure

1. Start with an initial interval $[a, b]$ such that $f(a)$ and $f(b)$ have opposite signs.
2. Compute the midpoint $c = \frac{a+b}{2}$.
3. Evaluate $f(c)$:
 - If $f(c) = 0$, then c is the root.
 - If $f(a) \cdot f(c) < 0$, then the root lies in $[a, c]$; set $b = c$.
 - If $f(c) \cdot f(b) < 0$, then the root lies in $[c, b]$; set $a = c$.
4. Repeat the process until the interval $[a, b]$ is sufficiently small or $|f(c)| < \epsilon$, where ϵ is the tolerance.

1.3 Pseudo Code

```
function BisectionMethod(f, a, b,  $\epsilon$ , N):  
    if  $f(a) * f(b) \geq 0$  then  
        return "Error:  $f(a)$  and  $f(b)$  must have opposite signs."  
  
    table = [] # To store iteration results  
    i = 1  
    while i <= N do  
        c = (a + b) / 2  
        fc = f(c)  
        table.append([i, a, b, c, f(a), f(b), fc])  
  
        if  $\text{abs}(fc) < \epsilon$  or  $(b - a) / 2 < \epsilon$  then  
            printTable(table)  
            return c # Root found  
  
        if  $f(a) * fc < 0$  then  
            b = c  
        else  
            a = c
```

```

        i = i + 1

    printTable(table)
    return "Failed to converge within the maximum number of iterations."

function printTable(table):
    print("Iter |    a    |    b    |    c    |    f(a)    |    f(b)    |    f(c)")
    for row in table do
        print(f"{row[0]:4} | {row[1]:8.5f} | {row[2]:8.5f} | {row[3]:8.5f} | {row[4]:8.5f} | {row[5]:8.5f} | {row[6]:8.5f}")

```

1.4 Sample Input and Output

Sample Input:

- Function: $f(x) = x^3 - x - 2$
- Interval: $[1, 2]$
- Tolerance: $\epsilon = 0.001$
- Maximum Iterations: $N = 20$

Sample Output:

Iter	a	b	c	f(a)	f(b)	f(c)
1	1.00000	2.00000	1.50000	-2.00000	4.00000	-0.12500
2	1.50000	2.00000	1.75000	-0.12500	4.00000	1.60938
3	1.50000	1.75000	1.62500	-0.12500	1.60938	0.66602
4	1.50000	1.62500	1.56250	-0.12500	0.66602	0.25293
5	1.50000	1.56250	1.53125	-0.12500	0.25293	0.06274
6	1.50000	1.53125	1.51563	-0.12500	0.06274	-0.03149
7	1.51563	1.53125	1.52344	-0.03149	0.06274	0.01562
8	1.51563	1.52344	1.51953	-0.03149	0.01562	-0.00781
9	1.51953	1.52344	1.52148	-0.00781	0.01562	0.00391
10	1.51953	1.52148	1.52051	-0.00781	0.00391	-0.00195
11	1.52051	1.52148	1.52099	-0.00195	0.00391	0.00098

Root found: 1.52099